

zkBugs Evaluation Results

Table 1: Tool Performance Summary

Tool	TP	FN	Timeout	Failure	Total	Median Time (s)
circumspect	1	55	0	0	56	0.08
circom_civer	1	13	14	28	56	120.81
picus	0	30	6	20	56	8.20
ecneproject	7	28	4	16	55	27.59
zkfuzz	4	12	22	18	56	20.69
conscs	1	22	25	8	56	11.43
TOTAL	14	160	71	90	335	11.11

Table 2: Execution Time Statistics

Tool	Total	Min (s)	Max (s)	Median (s)	Mean (s)	NR-Timeout
circumspect	52	0.01	29.68	0.08	1.55	0
circom_civer	14	0.06	552.42	120.81	205.57	14
picus	30	1.24	222.07	8.20	64.20	6
ecneproject	35	11.18	364.59	27.59	86.37	4
zkfuzz	16	0.01	59.47	20.69	19.59	22
conscs	23	0.27	476.22	11.43	62.28	25

Table 3: Bug-Tool Matrix (by Bug ID)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
1 FN	FN*	Timeout	FN	FN	Timeout
2 FN	FN	FN	FN	FN	TP*
3 FN	FN	FN	Failure	FN	TP*
4 FN	FN	Failure	FN	FN	Failure
5 TP*	FN	Failure	Failure	TP*	Failure
6 FN	FN*	Failure	FN	FN	Failure
7 FN	FN	TP	Failure	TP*	TP*
8 FN	FN	Failure	Failure	Failure	Failure
9 Timeout	FN	Timeout	FN	FN	Timeout
10 FN	FN	Timeout	FN	FN	Timeout
11 FN	TP	Timeout	FN	FN	Timeout
12 Timeout	FN	FN	Timeout	Failure	Timeout
13 Timeout	FN	FN	Failure	Failure	Failure
14 Timeout	FN	FN	Failure	Failure	Failure
15 Timeout	FN	Timeout	FN	FN	Timeout
16 Timeout	FN*	Failure	Failure	Timeout	Timeout
17 Timeout	FN*	Failure	Timeout	Failure	Failure
18 Failure	FN*	FN	Failure	Timeout	Failure
19 Failure	FN*	FN	Failure	Unknown	Failure
20	FN*	FN	Failure	Timeout	Failure

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Bug ID	circumspect	circom_civer	picus	ecneproject	zkfuzz
Failure 21	FN*	FN	Failure	Failure	Failure
Failure 22	FN	Failure	FN	FN	FN
Timeout 23	FN	Failure	FN*	Failure	FN*
Timeout 24	FN	Failure	FN	FN	FN
Timeout 25	FN	Failure	FN	FN	Timeout
FN 26	FN	Failure	FN*	Failure	FN*
Timeout 27	FN	Failure	FN*	FN*	FN*
Timeout 28	FN	FN	FN*	FN*	Timeout
FN 29	FN	Failure	FN*	FN*	FN
Timeout 30	FN	Timeout	Failure	TP*	Timeout
FN 31	FN	Timeout	FN	FN	TP*
FN 32	FN	FN	FN	FN	Timeout
Timeout 33	FN	Failure	FN	FN	FN
Timeout 34	FN	Failure	FN	FN	Timeout
FN 35	FN	Timeout	FN*	FN*	Timeout
FN 36	FN*	FN	FN	FN	Timeout
FN 37	FN*	FN	Failure	Timeout	Failure
Failure 38	FN	Timeout	Failure	FN	Timeout
FN 39	FN	Failure	FN	Failure	FN
Timeout					

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
40 FN	FN	Failure	FN	FN	FN
41 FN	FN	Failure	FN	FN	FN
42 Failure	FN*	Timeout	Failure	Failure	Failure
43 Timeout	FN	Failure	FN	Failure	FN
44 Timeout	FN	Failure	FN	FN	FN
45 FN	FN	Timeout	FN	FN	Timeout
46 FN	FN	Timeout	FN	FN	Timeout
47 FN	FN	Timeout	FN	FN	Timeout
48 Timeout	FN	Failure	Timeout	Failure	Failure
49 Timeout	FN	Failure	Timeout	Failure	Failure
50 Timeout	FN*	Failure	Failure	TP*	Timeout
51 Failure	FN	Failure	Failure	TP*	Timeout
52 Timeout	FN	Failure	Timeout	Failure	Failure
53 Failure	FN	Failure	Failure	TP*	Timeout
54 Timeout	FN*	Failure	Failure	TP*	Timeout
55 Timeout	FN	Timeout	Timeout	Failure	Failure
56 Timeout	FN	Failure	Failure	Failure	Failure

* = *manually analyzed*

Table 4: Bug ID to Name Mapping

ID	Bug Name
1	daira_hopwood_darkforest_v0_3_missing_bit_length_check
2	koukyosyumei_checkdepth_comparator_overflow
3	koukyosyumei_clamp_comparator_overflow
4	koukyosyumei_emitifinrange_lessthan_overflow
5	koukyosyumei_extractstringfrompoint_shiftedfirstbyte_unconstrained
6	koukyosyumei_spend_missing_range_check
7	koukyosyumei_subtract_unconstrained_multiplier
8	koukyosyumei_unconstrained_base64_decoded_len
9	koukyosyumei_under_constrained_date_encoder
10	leastauthority_Circuit_Does_Not_Check_the_ERC_20_Sum_Correctly_
11	leastauthority_previously_correct_ownership_proof_disabled_via_code_changes
12	matterlabs_email_spoofing_via_manipulated_from_header
13	matterlabs_emailauth_fails_with_overlapping_invitation_code_regex
14	matterlabs_fromaddrregex_allows_email_address_spoofing
15	matterlabs_underconstrained_fpmul_circuit
16	openzeppelin_mismatched_base64url_decoding_may_break_completeness
17	openzeppelin_non_determinism_of_some_inputs
18	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
19	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
20	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
21	veridise_arrayxor_is_under_constrained
22	veridise_babyjubjub_suborder_constraints_not_applied_correctly
23	veridise_blacklist_states_not_representable_in_field
24	veridise_bypassing_internal_transfer_limits_via_swaps
25	veridise_creditnoteproofs_can_be_stolen
26	veridise_data_escrow_encrypted_message_wrong_input
27	veridise_fortxreward_abstracted_away_in_reward_calc
28	veridise_incorrect_initialization_in_membership_circuits
29	veridise_kyt_signature_verification_fails_nonzero_hash
30	veridise_missing_range_checks_in_bigmod
31	veridise_missing_range_checks_on_comparison_circuits
32	veridise_no_zero_value_validation
33	veridise_nullifier_verification_can_be_disabled
34	veridise_ownershipproofs_could_identify_issuers
35	veridise_private_information_leakage
36	veridise_spammers_may_slash_themselves
37	veridise_template_CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
38	veridise_underconstrained_circuit_allows_invalid_comparison
39	veridise_unsafe_num2bits_254_blacklist_leaf

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ID	Bug Name
40	veridise_zaccount_renewal_kyc_expiry_not_validated
41	veridise_zaccount_renewal_multiple_nullifiers_same_utxo
42	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
43	veridise_zone_id_inclusion_prover_bypass
44	veridise_zone_limits_bypass_zswapv1
45	yacademy_Knowledge_of_any_member_signature_allow_to_generate_proof_of_membership
46	yacademy_input_signal_s_is_not_constrained_in_eff_ecdsa_circom
47	yacademy_under_constrained_circuits_compromising_the_soundness_of_the_system
48	zksecurity_an_attacker_can_craft_a_fake_non_inclusion_proof_for_a_given_key_due_to_an_aliasing_bug_in_the_smt_verifier
49	zksecurity_big_integer_zero_check_is_not_sound
50	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
51	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
52	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrary_L
53	zksecurity_missing_byte_range_checks_allows_packed_data_pollution
54	zksecurity_second_pre_image_attacks_on_packbytesandposeidon_may_be_used_to_register_arbitrary_passports_and_dsc_certificates
55	zksecurity_sha256_templates_return_zero_on_arbitrary_inputs
56	zksecurity_the_registration_and_disclosure_circuits_lack_range_checks_for_the_input_indices

Table 5: Execution Times per Bug (by Bug ID, in seconds)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
1 6.30	0.03	500.81	1.82	14.23	—
2 0.34	0.03	0.27	1.48	19.37	0.05
3 0.74	0.02	0.20	—	13.05	0.02
4 0.27	0.02	—	1.68	14.31	—
5 3.03	—	—	—	14.69	—
6 19.74	0.03	—	1.45	14.15	—
7 0.27	—	0.06	—	11.18	0.01
8 0.55	0.01	—	—	—	—
9 —	0.01	—	7.14	43.18	—
10 59.94	0.08	—	6.11	27.59	—
11 7.09	0.12	—	2.45	20.24	—
12 —	—	87.14	—	206.83	—
13 —	0.09	458.20	—	202.43	88.93
14 —	—	88.53	—	234.77	119.11
15 —	0.01	—	36.84	160.82	—
16 —	0.02	—	—	—	—
17 —	0.02	—	—	303.10	63.00
18 —	0.45	153.08	—	—	316.49
19 —	0.30	538.01	—	—	152.00
20	0.39	552.42	—	—	510.74

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
—					
21	0.28	382.08	—	274.24	582.87
—					
22	0.16	—	147.87	299.79	20.75
—					
23	0.13	—	148.66	152.89	24.19
—					
24	0.18	—	118.93	195.36	20.63
—					
25	0.02	—	9.25	55.83	—
146.63					
26	0.13	—	143.64	186.85	24.46
—					
27	0.18	—	202.28	345.87	22.04
—					
28	0.07	346.74	6.17	32.20	—
41.11					
29	0.14	—	222.07	333.07	17.64
—					
30	0.03	—	—	18.47	—
11.43					
31	0.01	504.04	1.24	13.43	0.36
4.69					
32	0.07	5.24	2.11	15.45	—
—					
33	0.14	—	218.33	364.59	21.46
—					
34	0.09	—	1.58	18.88	—
0.31					
35	0.08	—	11.03	40.65	—
91.44					
36	0.09	3.20	1.69	14.27	—
1.13					
37	0.50	262.77	—	—	166.88
—					
38	0.02	536.84	—	13.54	—
34.16					
39	0.13	—	182.36	206.56	21.98
—					

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
40	0.12	—	40.45	90.91	59.47
476.22					
41	0.20	—	24.45	68.21	41.47
475.47					
42	0.30	—	—	543.44	139.74
—					
43	0.15	—	163.60	216.89	20.53
—					
44	0.14	—	211.95	363.60	18.42
—					
45	0.02	—	3.25	23.91	—
17.29					
46	0.01	—	3.09	23.27	—
17.20					
47	0.01	—	3.12	23.28	—
17.03					
48	29.68	—	—	476.77	121.23
—					
49	16.58	—	—	280.75	106.18
—					
50	0.04	—	—	100.38	—
—					
51	0.03	—	—	62.04	—
557.61					
52	14.71	—	—	428.08	173.59
—					
53	0.02	—	—	60.00	—
506.93					
54	0.04	—	—	92.98	—
—					
55	0.03	—	—	211.91	80.72
—					
56	14.28	—	—	284.53	180.84
—					